

Dr. Ashley N. Ives

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Citizenship: United States

SUMMARY

Ph.D. in chemistry with over 10 years of experience in mass spectrometry-based proteomics. My doctoral work culminated in three first-author publications (*JACS*, *Anal. Chem.*, *JASMS*), two NIH grants (F31, T32), and spanned a broad range of wet lab techniques including cell culture, protein purification, and recombinant DNA methods — giving me a strong and versatile foundation at the bench. Since joining Pacific Northwest National Laboratory as a postdoctoral researcher, my focus has shifted toward statistical analysis and interpretation of large-scale proteomics datasets. I have thrived in a fast-paced, highly collaborative environment where I contribute to many diverse projects simultaneously, and this breadth of work has sharpened my ability to communicate complex scientific concepts to both specialist and non-specialist audiences. Overall, I have extensive experience with all facets of proteomics including native and denatured top-down, single-cell proteomics, traditional bottom-up, and selected reaction monitoring.

EDUCATION

Postdoctoral Research Associate

Pacific Northwest National Laboratory, Richland, WA

August 2023–Present

Doctorate of Philosophy in Chemistry

Northwestern University, Evanston, IL

July 2023

Bachelor of Science in Chemistry

University of California, Berkeley, CA

May 2017

PUBLICATIONS

1. **Ives AN**, Thibert S, Berger MR, Gafrey MJ, Mitchell HD, Williams SM, Zhou M, Waters KM, Sims AC, Zhang T. “Human Coronavirus 229E Infection Alters Histone Proteoforms” *J Proteome Res.* **2026**, Accepted.
2. **Ives AN**, Fulcher JM, Bautista AR, Kitata RB, Williams SM, Bennett DA, De Jager PL, Petyuk VA. “Single-Cell Nanodroplet Processing Proteomics Pipeline for Analysis of Human-Derived Microglia” *Proteomics.* **2026**, e70112.
3. Li X, Kabza A, **Ives AN**, Thiel J, Waters KM, Qian WJ, Sims AC, Zhang T. “Proteome-wide characterization of PTMs reveals host cell responses to viral infection and identifies putative antiviral drug targets” *Front Immunol.* **2025**, 16, 1587106.
4. Uychoco P, Majorek KA, **Ives AN**, Le VTB, Caro De Silva PL, Paurus VL, Attah IK, Lipton MS, Minor W, Kuhn ML. “Structural, functional, and regulatory evaluation of a cysteine post-translationally modified Gcn5-related N-acetyltransferase” *Biochem Biophys Res Commun.* **2025**, 748, 151299.
5. **Ives AN**, Jooß K, Melani RD, Fellers RT, Janetzko J, Kelleher NL. “Capillary Zone Electrophoresis-Mass Spectrometry of Intact G Protein-Coupled Receptors Enables Proteoform Profiling” *Anal Chem.* **2025**, 97(13), 7307–7316.
6. Fulcher JM, **Ives AN**, Tasaki S, Kelly SS, Williams SM, Fillmore TL, Zhou M, Moore RJ, Qian WJ, Paša-Tolić L, Yu L, Oveisgharan S, Bennett DA, De Jager PL, Petyuk VA. “Discovery of Proteoforms Associated With Alzheimer’s Disease Through Quantitative Top-Down Proteomics” *Mol Cell Proteomics.* **2025**, 24(6), 100983.
7. **Ives AN**, Sagendorf TJ, Nierves L, Lin TT, Dirice E, Kulkarni RN, Paša-Tolić L, Qian WJ, Fulcher JM. “Characterization of Cytokine Treatment on Human Pancreatic Islets by Top-Down Proteomics”

Proteomics. **2025**, 25(24), 55–66.

8. **Ives AN**, Dunn HA, Afsari HS, Dos Santos Seckler H, Foroutan MJ, Chavez E, Melani RD, Fellers RT, LeDuc RD, Thomas PM, Martemyanov KA, Kelleher NL, Vafabakhsh R. “Middle-Down Mass Spectrometry Reveals Activity-Modifying Phosphorylation Barcode in a Class C G Protein-Coupled Receptor” *J Am Chem Soc*. **2022**, 144(50), 23104–23114.
9. **Ives AN**, Su T, Durbin KR, Early BP, Dos Santos Seckler H, Fellers RT, LeDuc RD, Schachner LF, Patrie SM, Kelleher NL. “Using 10,000 Fragment Ions to Inform Scoring in Native Top-down Proteomics” *J Am Soc Mass Spectrom*. **2020**, 31(7), 1398–1409.
10. Kafader JO, Melani RD, Schachner LF, **Ives AN**, Patrie SM, Kelleher NL, Compton PD. “Native vs Denatured: An in Depth Investigation of Charge State and Isotope Distributions” *J Am Soc Mass Spectrom*. **2020**, 31(3), 574–581.
11. Schachner LF, **Ives AN**, McGee JP, Melani RD, Kafader JO, Compton PD, Patrie SM, Kelleher NL. “Standard Proteoforms and Their Complexes for Native Mass Spectrometry” *J Am Soc Mass Spectrom*. **2019**, 30(7), 1190–1198.
12. Cline EN, Das A, Bicca MA, Mohammad SN, Schachner LF, Kamel JM, DiNunno N, Weng A, Paschall JD, Bu RL, Khan FM, Rollins MG, **Ives AN**, Shekhawat G, Nunes-Tavares N, de Mello FG, Compton PD, Kelleher NL, Klein WL. “A novel crosslinking protocol stabilizes amyloid β oligomers capable of inducing Alzheimer’s-associated pathologies” *J Neurochem*. **2019**, 148(6), 822–836.
13. Roberts AM, Miyamoto DK, Huffman TR, Bateman LA, **Ives AN**, Akopian D, Heslin MJ, Contreras CM, Rape M, Skibola CF, Nomura DK. “Chemoproteomic Screening of Covalent Ligands Reveals UBA5 As a Novel Pancreatic Cancer Target” *ACS Chem Biol*. **2017**, 12(4), 899–904.

GRANTS AND SCHOLARSHIPS

Ruth L. Kirschstein Predoctoral Research Service Award (F31)	02/2022–01/2023
F31MH129114-01	

Chemistry of Life Processes Fellowship (T32)	08/2018–08/2020
NIGMS Chemistry-Biology Interface Training Program supported by T32GM105538	

SKILLS

Programming: R (RStudio) and Python.

Mass Spectrometry: Sample preparation for bulk or single-cell bottom-up and top-down proteomics; DDA and DIA method development including targeted fragmentation optimization (higher collisional dissociation, electron-transfer dissociation, proton transfer reaction, etc.); micro- and nanoflow liquid chromatography; native and denatured capillary electrophoresis; native proteomics sample preparation and method development.

Mass Spectrometry Data Analysis: Proficient in many proteomics database search engines including DiaNN, Spectronaut, MSFragger (FragPipe), SkyLine, TopPIC, and UniDec; extensive experience with data analysis and figure generation with R (RStudio).

Recombinant DNA Methods: Gibson assembly, restriction cloning, QuikChange site-directed mutagenesis, PCR, sequencing and interpreting sequencing results, DNA electrophoresis, plasmid DNA extraction, bacterial transformation.

Molecular Biology: Immunoprecipitation and purification of soluble and membrane proteins, enzyme and signaling kinetics via luciferase-based assays, protein electrophoresis (SDS- and Native PAGE) and immunostaining, labeling and sorting cell suspensions via cellenONE single-cell dispensers.

Confocal Microscopy: Live cell imaging using Zeiss instrumentation; cell surface and permeable labeling of protein targets for fluorescence microscopy; image workup and quantification using Fiji.

Cell Culture: Mammalian and bacterial cell culture, lentiviral transduction, bacterial protein expression, chemical transfection.

Document Creation: Quarto, Microsoft Office Suite, Adobe Illustrator, BioRender, Snapgene, L^AT_EX.

General: Detailed record-keeper, organized, great communicator, strong interpersonal skills and good team-player.

WORK EXPERIENCE

Postdoctoral Research Associate

08/2023–Present

Pacific Northwest National Laboratory, Richland, WA

Advisors: Drs. Vladislav Petyuk, James Fulcher

- Developed a single-cell proteomics platform to characterize single microglia and single neurons from human donor samples.
- Developed a high-throughput method to characterize metal-binding proteins using native mass spectrometry in a 96-well format.
- Performed numerous top-down and bottom-up proteomics studies to support the EMSL user program; includes both plant and mammalian proteomics experiments.

Graduate Research Assistant

06/2017–07/2023

Chemistry Department, Northwestern University, Evanston, IL

Advisors: Drs. Neil Kelleher and Reza Vafabakhsh

- Characterized phosphorylation states of G protein-coupled receptors using both middle-down and top-down platforms.
- Developed stable mammalian cell lines and protein purification protocols.
- Optimized capillary zone electrophoresis separations for peptide and intact protein separations.

Undergraduate Researcher

08/2014–05/2017

Chemistry Department, University of California, Berkeley, CA

Advisor: Dr. Daniel Nomura

- Performed mammalian cell culture and bottom-up mass spectrometry and analysis.
- Performed activity-based protein profiling to identify potential drug candidates from heterologous cell lines.

HONORS AND AWARDS

Steven A. Hofstadler Graduate Student Award

2019

Presented at Clinical and Pharmaceutical Solutions through Analysis Annual Meeting for outstanding abstract and session presentation.

PRESENTATIONS

1. Oral presentation. 16th Annual Rush Alzheimer's Disease Center ROSMAP Investigator's Meeting. Rush University Medical Center, Chicago, IL. May 20, 2025. **Ives AN**, Fulcher JM, Petyuk V. "Single-cell proteomics of human-derived microglia using nanodroplet processing (nanoPOTs)".
2. Oral presentation. American Society of Mass Spectrometry Annual Meeting. Anaheim, CA. June 5, 2024. **Ives AN**, Fulcher JM, Petyuk V. "Mapping Proteoforms Associated with Alzheimer's Disease Through Quantitative Shot-Gun Top-Down Proteomics of 103 Human Patient Samples".
3. Oral presentation. American Society of Mass Spectrometry Annual Meeting. Minneapolis, MN. June 8, 2022. **Ives AN**, Dos Santos Seckler H, Kelleher NL, Vafabakhsh R. "Resolving isomeric phosphorylation states of GPCR C-termini using capillary electrophoresis (CE-MS/MS)".
4. Oral presentation. American Society of Mass Spectrometry Annual Meeting. Virtual. June 3, 2020. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. "Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry".
5. Oral presentation. Clinical and Pharmaceutical Solutions through Analysis Annual Meeting. Langhorne, PA. October 30, 2019. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie

- SM, Kelleher NL. “Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry”.
6. Oral presentation. Chicago Mass Spectrometry Day. University of Chicago, Chicago, IL. July 19, 2019. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. “Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry”.
 7. Poster presentation. ASBMB 15th International Symposium on Proteomics in the Life Sciences. Boston, MA. August 17, 2025. **Ives AN**, Fulcher JM, Petyuk V. “Single-cell proteomics of human-derived microglia using nanodroplet processing (nanoPOTs)”.
 8. Poster presentation. US Human Proteome Organization Annual Meeting. Chicago, IL. March 7, 2023. **Ives AN**, Dunn HA, Afsari HS, Dos Santos Seckler H, Foroutan MJ, Chavez E, Melani RD, Fellers RT, LeDuc RD, Thomas PM, Martemyanov KA, Kelleher NL, Vafabakhsh R. “Middle-Down Mass Spectrometry Reveals Activity-Modifying Phosphorylation Barcode in a Class C G Protein-Coupled Receptor”.
 9. Poster presentation. American Society of Mass Spectrometry Annual Meeting. Atlanta, GA. June 4, 2019. **Ives AN**, Su T, Dos Santos Seckler H, Schachner LF, McGee J, Fellers RT, Patrie SM, Kelleher NL. “Trends from >15,000 Assigned Fragment Ions in Top-Down Mass Spectrometry”.
 10. Poster presentation. Keystone Symposium: G Protein-Coupled Receptors. Salt Lake City, UT. April 7, 2022. **Ives AN**, Dunn HA, Afsari HS, Dos Santos Seckler H, Foroutan MJ, Chavez E, Melani RD, Fellers RT, LeDuc RD, Thomas PM, Martemyanov KA, Kelleher NL, Vafabakhsh R. “Middle-Down Mass Spectrometry Reveals Activity-Modifying Phosphorylation Barcode in a Class C G Protein-Coupled Receptor”.

TEACHING AND MENTORSHIP

Northwestern Summer Research Opportunity Program Mentor 2019, 2022
 Trained and mentored undergraduates in laboratory work and scientific presentation for eight weeks. Developed a project for each student and provided technical training as well as professional support and development.

OUTREACH

Letters to a Pre-Scientist Participant 2018–2020
 Served as a pen pal with middle school students. The focus is to build personal connections and help demystify science for students from high-poverty schools.

REFERENCES

Tong Zhang, PhD	Chemist tong.zhang@pnml.gov (509) 371-7780	Pacific Northwest National Laboratory Biological Sciences Division, Richland, WA
James Fulcher, PhD	Chemist james.fulcher@pnml.gov	Pacific Northwest National Laboratory Environmental Molecular Sciences Laboratory, Richland, WA
Vladislav Petyuk, PhD	Data Scientist vladislav.petyuk@pnml.gov	Pacific Northwest National Laboratory Biological Sciences Division, Richland, WA